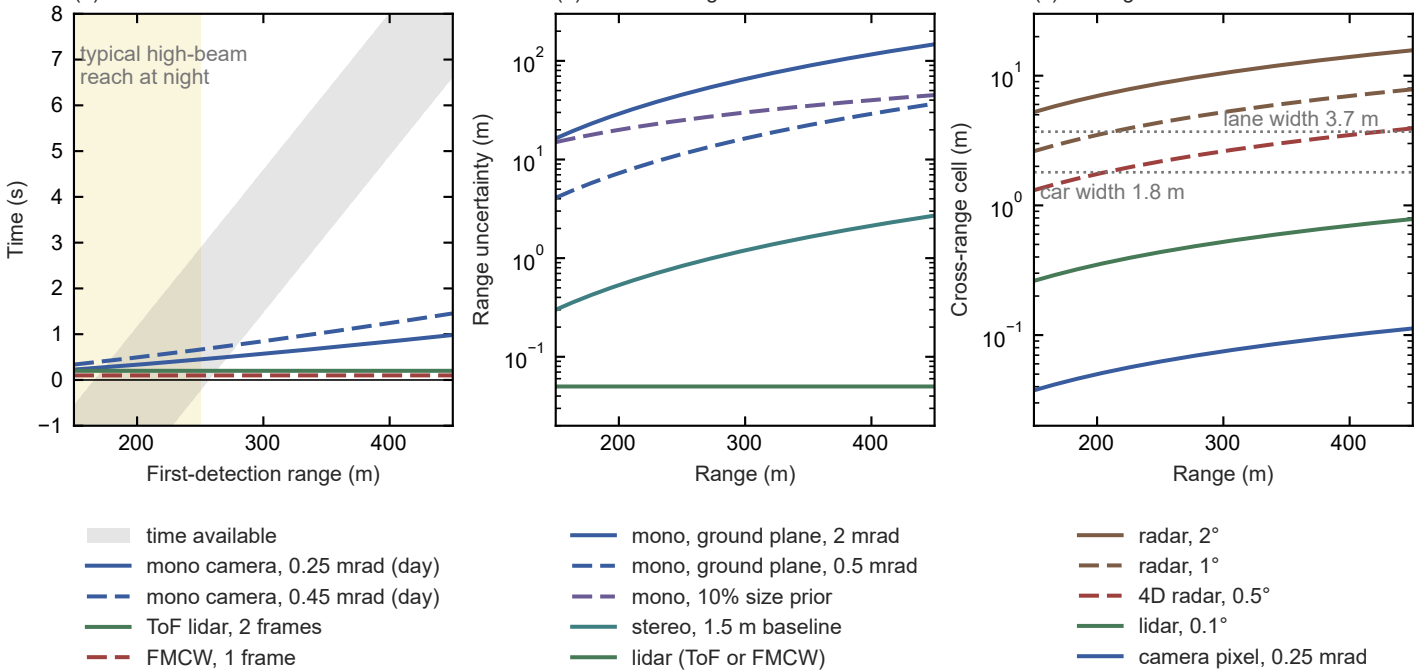
(a) Time to decide vs time available
typical high-beam reach at night
Time (s)
First-detection range (m)
(b) Absolute range error
Range uncertainty (m)
Range (m)
(c) Placing a static return in-lane
Cross-range cell (m)
lane width 3.7 m
car width 1.8 m
Range (m)
time available
mono camera, 0.25 mrad (day)
mono camera, 0.45 mrad (day)
ToF lidar, 2 frames
FMCW, 1 frame
mono, ground plane, 2 mrad
mono, ground plane, 0.5 mrad
mono, 10% size prior
stereo, 1.5 m baseline
lidar (ToF or FMCW)
radar, 2°
radar, 1°
4D radar, 0.5°
lidar, 0.1°
camera pixel, 0.25 mrad